



BRAC UNIVERSITY
Department of Computer Science and Engineering
M.Sc. in Computer Science and Engineering Program
Final Exam, Fall 2025
CSE710 (Advanced Artificial Intelligence), Section-1

Full Marks: 30
Time: 120 minutes

Note: There are three questions; answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned in the right margin.

1. True emotions are difficult to predict. Even a smiling face can cover the boundless grief. However, to predict true emotion, we can analyze psycho-physiological signals collected through the non-invasive Internet of Medical Things. By analyzing psycho-physiological signals, we will get the observations of Valence and Arousal. Valence and Arousal can be positive or negative. Based on the output of Valence and Arousal, we can determine true emotion using the emotion wheel (See Figure 1). [CO2, C3, Marks: 6+2+2]

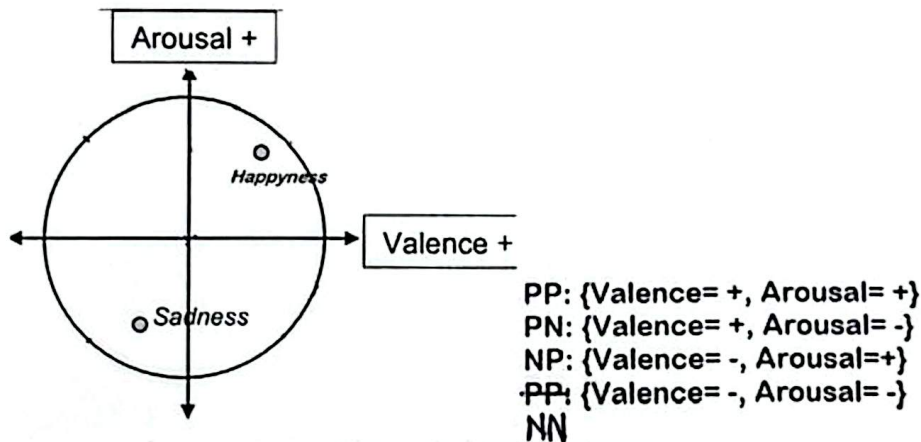


Figure 1: Emotion Wheel

For that purpose, we want to build an HMM model for predicting hidden emotions, namely 'Happy' and 'Sad', from the Valence and Arousal observations. The 3-hour observations of a single day are as follows:

Table -1: Observation sequences

	1st-hour	2nd-hour	3rd-hour		
day 1	PP	NN	PN		

Considering the given initial parameter values in Figure 2 and observation sequences in Table 1:

- Determine the Expected number of transitions from state *Happy* to state *Sad* at time 1.
- Determine the Expected number of times for being in state *Happy* at time 1.
- Determine the initial probability of *Happy* and *Sad* states.

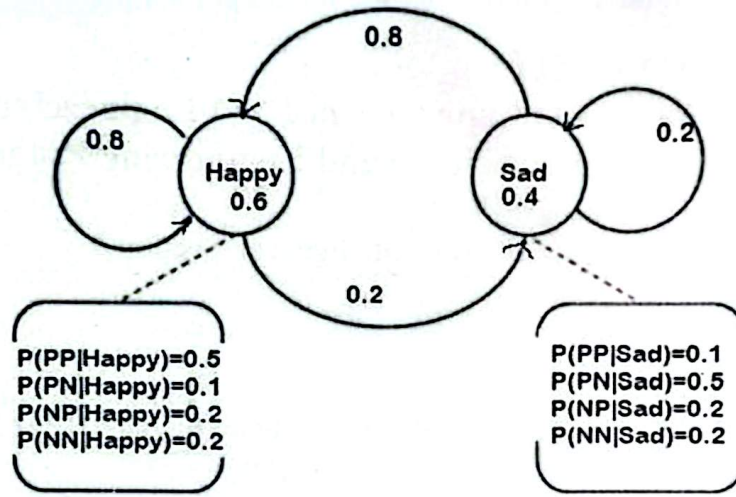


Figure 2: HMM model for Emotion Recognition

2. Consider a sample dataset $X = \{x_1, x_2, x_3, x_4, x_5\} = \{1, 2, 3, 4, _ \}$ from a [CO1, C3, Marks: 10]
population of Normal or Gaussian distribution with missing data x_5 . We want to estimate the population mean (μ) and standard deviation (σ) from the sample data using Maximum Likelihood Estimation (MLE). As the dataset has a missing data, you have to use the Expectation Maximization (EM) algorithm to determine the population mean (μ) and standard deviation (σ). In the initial E-step, you have to estimate the x_5 by averaging all the values of the set X . However, from the 2nd iteration, in the E-step, you have to predict the missing value using the following equation for z-score determination.

$x_5 = Z * \sigma + \mu$; considering $Z=2$. In the M-step, you have to compute the MLE for estimating μ and σ . The log-likelihood of a Gaussian distribution is as follows:

$$\text{Log}(X | \mu, \sigma) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2$$

Where n is the number of elements in X . Infer the value of μ and σ within the range 1 to 3 by assuming μ and σ as integer variables and by estimating the Maximum Log-Likelihood of the Gaussian distribution following the EM algorithm up to 2 iterations. Predict the missing value after the 2nd iteration.

3. Proof that the expected number of transitions from state i to state j at time t is:

[CO3, C3, Marks: 5+5]

$$\xi_t(i, j) = \frac{\alpha_t(i) a_{ij} b_j(z_{t+1}) \beta_{t+1}(j)}{\sum_{k,l} \alpha_t(k) a_{kl} b_l(z_{t+1}) \beta_{t+1}(l)}$$

Where, $Z_{1:T}$ is the observation sequence, a_{ij} is the transition probability, $b_j()$ is the emission probability, $\alpha()$ is the forward probability and $\beta()$ is the backward probability. Also, derive the probability of the hidden state at time t is k given the observation sequence $Z_{1:T}$ i.e.,

$$p(S_t = k | Z_{1:T}) = \frac{\alpha_t^k \beta_t^k}{\sum_i \alpha_t^i \beta_t^i}$$